

Experiment-5

PRACTICE QUERIES - I

AIM: Practice Queries.

Step-1: Write a Query to show the table employee_lavina.

```
mysql> select * from employee_lavina;
```

employee_id	first_name	last_name	salary	department_id	hire
1	John	Smith	60000	1	2022-01-15
2	Jane	Doe	15000	1	2021-03-20
3	Peter	Jones	90000	2	2020-05-10
4	Mary	Williams	80000	2	2021-08-01
5	David	Brown	65000	3	2023-10-05
6	null	Davis	55000	3	2023-02-12

6 rows in set (0.00 sec)

Step-2: Write a Query to show the table departments.

```
mysql> select * from departments;
```

department_id	department_name
1	Engineering
2	Sales
3	Marketing
4	HR

4 rows in set (0.00 sec)

Step-3: Write a SQL query to create a new column Salary_bracket characterised each employee based on their salary.

```
mysql> select first_name, salary,  
-> CASE  
-> WHEN salary > 85000 THEN 'higher earner'  
-> WHEN salary > 70000 THEN 'mid earner'  
-> else 'standard earner'  
-> end as salary_bracket  
-> from employee_lavina;
```

first_name	salary	salary_bracket
John	60000	standard earner
Jane	75000	mid earner
Peter	90000	higher earner
Mary	80000	mid earner
David	65000	standard earner
NULL	55000	standard earner

6 rows in set (0.01 sec)



Step-4: Write a Query to find the name of all the employee who works in "Sales" department.

Method-1:

```
mysql> select first_name, last_name
-> from employee_lavina as E, departments as D
-> where E.department_id = D.department_id and D.department_name = "Sales";
```

first_name	last_name
Peter	Jones
Mary	Williams

2 rows in set (0.00 sec)

Method-2:

```
mysql> select first_name, last_name
-> from employee_lavina
-> where department_id IN (select department_id
-> from departments
-> where department_name = "sales");
```

first_name	last_name
Peter	Jones
Mary	Williams

2 rows in set (0.01 sec)

Step-5: Write a SQL statement to display employee_name, Salary of the employee who earn less than 30000.

```
mysql> select first_name, last_name, salary
-> from employee_lavina
-> where salary < 30000;
```

Empty set (0.00 sec)

Step-6: Write a SQL statements to display name and salary of each employee who Earn between 55000 and 75000.

```
mysql> select first_name, last_name, salary
-> from employee_lavina
-> where salary between 55000 and 75000;
```

first_name	last_name	salary
John	Smith	60000
Jane	Doe	75000
David	Brown	65000
NULL	David	55000

4 rows in set (0.00 sec)

Step-7: Write a SQL query to display the annual salary of each employee display the annual salary column as annual salary.

```
mysql> select employee_id, first_name, last_name, 12*salary as "Annual salary"
-> from employee_lavina;
```

employee_id	first_name	last_name	Annual salary
1	John	Smith	720000
2	Jane	Doe	900000
3	Peter	Jones	1080000
4	Mary	Williams	960000
5	David	Brown	780000
6	NULL	David	660000

6 rows in set (0.03 sec)

Step-8: Write a SQL query for finding the employees who were hired in the year 2023 and month January (01).

```
mysql> select *
-> from employee_lavina
-> where year(hire) = 2023 and month(hire) = 01;
```

employee_id	first_name	last_name	salary	department_id	hire
5	David	Brown	65000	3	2023-01-05

1 row in set (0.01 sec)

Step-9: Write a SQL query to display the name and the salary of an employee whose second letter of the first name is 'a'.

```
mysql> select first_name, salary
-> from employee_lavina
-> where first_name like '_a%';
```

first_name	salary
Jane	75000
Mary	80000
David	65000

3 rows in set (0.01 sec)

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Practice Queries- II

AIM: Practice Queries.

Step -1: Create table Publisher_lavina with given details.

```
mysql> select * from publisher_lavina;
```

p_code	p_name	p_city	p_state
AH	Arham publ	Udaipur	RJ
AP	Wilson publ	newyork	NY
AM	jerry publ	newyork	NY
BB	arihant publ	jaipur	RJ
BH	best publ	udaipur	RJ

5 rows in set (0.00 sec)

Step-2: create table Author_lavina with given details.

```
mysql> select * from author_lavina;
```

a_number	a_first	a_last
1	raghu	raman
2	jai	narayan
3	clarke	agasthe
4	archer	francis
5	clive	jeffrey

5 rows in set (0.00 sec)

Step-3: create table Book_lavina with the given details.

```
mysql> select * from book_lavina;
```

b_code	b_title	p_code	b_type	b_price
100	DBMS	BB	SUS	1200
200	DCCN	BB	SUS	500
300	DCCN	BB	SUS	500
400	CPP	BB	LANG	1000
500	TOC	BB	LANG	1300

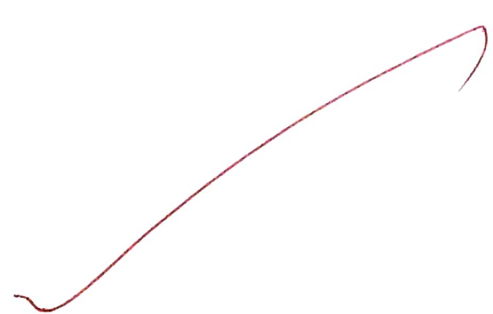
5 rows in set (0.01 sec)

Step-4: create table Wrote_lavina with the given details.

```
mysql> select * from wrote_lavina;
```

b_code	a_number	seq_number
100	1	1
100	2	1
200	2	1
100	4	1
6328	5	1
7484	6	1

6 rows in set (0.03 sec)

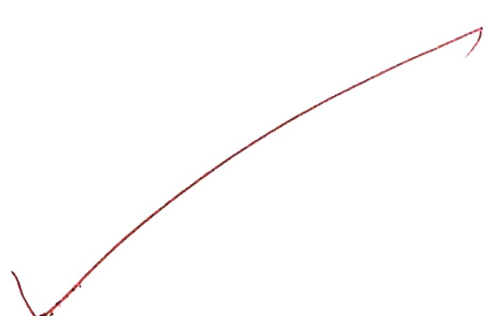


Step-5: Find the average price of all the books which have SUS as their book type.

```
mysql> select AVG(b_price) as price
-> from book_lavina
-> where b_type = "SUS";
```

price
733.3333

1 row in set (0.01 sec)



Step-6: List the name of every Publisher not located in New York state.

```
mysql> select p_name
-> from publisher_lavina as P
-> where p_city != "newyork";
```

p_name
Arham publ
arihant publ
best publ

3 rows in set (0.00 sec)



Step-7: List the book code and title of every book whose type is "LANG" or "SYNC". Use the IN operator in your command.

```
mysql> select b_code, b_title
-> from book_lavina B
-> where b_type IN ("LANG","SYNC");
```

b_code	b_title
400	CPP
500	TOC

2 rows in set (0.00 sec)

Step-8: Determine the name of the most expensive and cheaper book given to you .

```
mysql> SELECT b_title, b_price
-> FROM book_lavina
-> WHERE b_price = (SELECT MAX(b_price) FROM book_lavina)
-> OR b_price = (SELECT MIN(b_price) FROM book_lavina);
```

b_title	b_price
DCCN	500
DCCN	500
TOC	1300

3 rows in set (0.06 sec)

Step-9: Find the book title for every book written by author number '01'. Use the IN operator in your format.

```
mysql> SELECT b_title
-> FROM book_lavina
-> WHERE b_code
-> IN (SELECT b_code FROM wrote_lavina WHERE a_number =1);
```

b_title
DBMS

1 row in set (0.03 sec)

Experiment-6 Practice Query III

AIM: Practice queries.

Step-1: create table student.

```
mysql> select * from students;
```

student_id	student_name
9090	Jane Eyre
9540	Sherlock Holmes
9021	Raven
9632	Annabel lee
9255	van Helsing

5 rows in set (0.00 sec)

Step-2: create table Professors.

```
mysql> select * from Professors;
```

Professor_id	Professor_name
10445	Broke
13543	Doyle
12450	Poe
13085	Stoker

4 rows in set (0.00 sec)

Step-3: create table Grades.

```
mysql> select * from Grades;
```

student_id	Professor_id	grade
9090	10445	A
9540	13543	B
9021	12450	A
9632	12450	B
9255	13085	C

5 rows in set (0.00 sec)

Step-4: Write the name of the Student whose name start with J.

```
mysql> select student_name  
-> from students  
-> where student_name like "J%";
```

student_name
Jane Eyre

1 row in set (0.01 sec)

Step 5: Write the student name that ends with alphabet n.

```
mysql> select student_name
-> from students
-> where student_name like "%n";
+-----+
| student_name |
+-----+
| Raven        |
+-----+
1 row in set (0.00 sec)
```

Step-6: Find the name of the student who have professor "Broke" assigned to them.

```
mysql> select student_name
-> from students s, Professors p, Grades g
-> where s.student_id= g.student_id and p.Professor_id= g.Professor_id and p.Professor_name="broke";
+-----+
| student_name |
+-----+
| Jane Eyre    |
+-----+
1 row in set (0.00 sec)
```

Step-7: Find the student who scored Grade 'A'.

```
mysql> select student_name
-> from students s, Grades g
-> where s.student_id = g.student_id and g.grade = "A";
+-----+
| student_name |
+-----+
| Jane Eyre    |
| Raven        |
+-----+
2 rows in set (0.00 sec)
```

Step-8: Find the Total number of students who scored Grade 'B'.

```
mysql> select count(*)
-> from Grades g
-> where g.grade = "B";
+-----+
| count(*) |
+-----+
|         2 |
+-----+
1 row in set (0.01 sec)
```

Step-9: Find the Total count of the students for each grade.

```
mysql> select grade, count(student_id)
-> from Grades group by grade;
```

grade	count(student_id)
A	2
B	2
C	1

3 rows in set (0.01 sec)

Step-10: Find the student who have professor either Stoker or Poe.

```
mysql> select *
-> from Professors
-> where Professor_name in ('Stoker', 'Poe');
```

Professor_id	Professor_name
12450	Poe
13085	Stoker

2 rows in set (0.01 sec)

Step-11: Find the name of the Students who have 5 alphabet in their name.

```
mysql> select student_name
-> from students
-> where student_name like "_____";
```

student_name
Raven

1 row in set (0.00 sec)

Step-12: Find the name of the student who have atleast 7 character in their name.

```
mysql> select student_name
-> from students
-> where student_name like "_____";
```

student_name
Jane Eyre
Sherlock Holmes
Annabel lee
van Helsing

```
4 rows in set (0.00 sec)
```

Step-13: Find the name who have alphabet "H" in the middle of their name.

```
mysql> select student_name
-> from students
-> where student_name like "%H%";
```

student_name
Sherlock Holmes
van Helsing

```
2 rows in set (0.00 sec)
```